

6: Microbiology and Pathogens

Edexcel B A-level Biology (9BI0), Topic 6: growing, counting and isolating bacteria; how bacteria, a fungus, a virus and a protoctist cause disease; antibiotics and resistance; controlling malaria; and the immune response and vaccination.

6.1 Microbial Techniques

6.1 (i) Aseptic Technique

Aseptic technique: A set of working practices that stop unwanted microorganisms from getting into a culture (contamination) and stop the cultured microorganisms from escaping into the environment or infecting the person doing the work.

Even a harmless strain is handled as if it could be a pathogen, because a culture can be contaminated from the air, the bench or the worker's skin. The standard steps, and the reason for each, are:

- **Disinfect the bench** before and after the work, to kill microorganisms on the surface.
- **Work close to a lit Bunsen burner.** The rising hot air carries airborne microorganisms and dust up and away from open cultures.
- **Sterilise equipment and media before use** in an autoclave (steam under pressure, typically 121 °C for about 15 minutes), which kills all microorganisms including bacterial spores.
- **Flame the wire inoculating loop** until it glows red before and after every transfer, then let it cool so it does not kill the cells being transferred.
- **Flame the neck of a culture bottle** immediately after removing the lid and again before replacing it. This kills microorganisms on the rim and creates an upward air current out of the bottle.
- **Open lids as little as possible**, lifting a Petri dish lid only at an angle, to reduce entry of airborne microorganisms.

- **Secure, but do not fully seal, plates.** Two short strips of tape hold the lid on. Sealing all the way round would allow anaerobic conditions to develop, which favours some dangerous anaerobes.
- **Incubate at no more than about 25 °C** in a school laboratory. Many human pathogens are adapted to grow best near body temperature (37 °C), so a lower temperature reduces the risk of growing them.
- **Dispose of cultures safely:** do not open plates after incubation, autoclave everything before disposal, and wash hands.

Exam Tip: Questions on aseptic technique usually carry a mark for the step and a mark for the reason. "Flame the loop" earns little on its own; "flame the loop until red hot to kill any microorganisms on it, and allow it to cool so that the bacteria being transferred are not killed" earns both. Keep two purposes separate in your answer: preventing contamination of the culture, and protecting the worker and the environment.

6.1 (ii) Principles of Culturing Microorganisms

Culturing means providing the conditions a microorganism needs to grow and reproduce. Bacteria increase in number by **binary fission**, in ideal conditions as often as every 20–30 minutes. They need:

- **Nutrients and water:** a carbon and energy source (often glucose), a nitrogen source for amino acids and nucleotides (such as peptones or yeast extract), mineral ions and sometimes vitamins.
- **A suitable temperature and pH:** both affect enzyme activity. Too cold and growth is slow; too hot, or the wrong pH, and enzymes denature. Buffers may be added to the medium.
- **Oxygen, or its absence:** obligate aerobes need oxygen; obligate anaerobes are killed by it; facultative anaerobes grow either way. Liquid cultures of aerobes are often shaken or stirred to keep them aerated.

The general procedure is to prepare and sterilise a medium, **inoculate** it with a small number of cells using aseptic technique, then **incubate** it at a controlled temperature for a set time. In a closed container with no nutrients added and no waste removed, this is a **batch culture**, and the population follows the growth curve described in 6.1 (v).

6.1 (iii) Types of Culture Media

Medium	What it is	Typical use
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Broth (liquid) culture	A sterile solution of nutrients, such as nutrient broth, in a flask, bottle or test tube	Growing large numbers of cells; following population growth over time by taking samples (turbidity, dilution plating)
Agar (solid) medium	Nutrient broth set with agar, a seaweed polysaccharide that stays a gel at incubation temperatures and that most microorganisms cannot digest	Growing separate colonies to count them (viable counts) or isolate them (streak plating)
Selective medium	A broth or agar containing a substance that stops some microorganisms growing while allowing others to grow, or a nutrient that only certain organisms can use	Isolating one type from a mixture, for example agar containing an antibiotic so that only cells carrying a resistance gene form colonies, or high-salt agar that most bacteria cannot tolerate but staphylococci can

6.1 (iv) Measuring the Growth of a Bacterial Culture

There are four main approaches. The key distinction is between methods that count **all cells** (a total count) and methods that count only **living cells** (a viable count).

Method	How it works	Counts	Advantages / limitations
Cell count (haemocytometer)	Cells in a counting chamber of known volume are counted under a microscope	Total (live and dead)	Quick and direct; bacteria are small and hard to count; small sample
Dilution plating (viable count)	The culture is serially diluted and spread on agar; each colony is assumed to come from one living cell	Living cells only	Counts only viable cells; slow (24–48 h); clumped cells give one colony, so it can underestimate
Mass	Cells from a known volume are centrifuged or filtered, then weighed wet or dried to constant mass	Total biomass	Useful for fungi and large cultures; needs a large sample; drying kills the cells

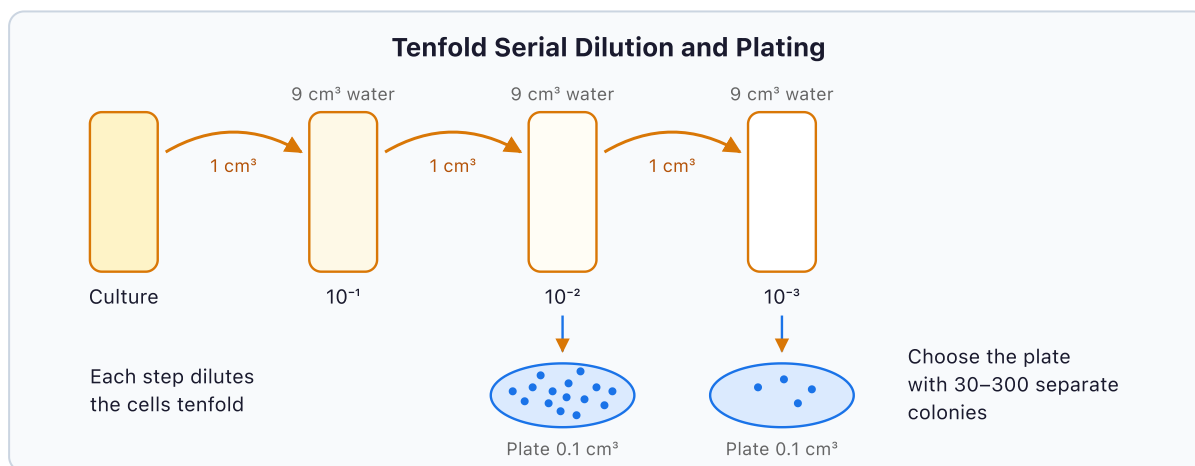
Optical method (turbidity)	A colorimeter, zeroed with sterile broth, measures absorbance, which rises as more cells make the broth cloudier; a calibration curve converts readings to cell numbers	Total (live and dead)	Fast and can be repeated often; cannot tell live cells from dead ones
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Serial Dilution and Calculating a Viable Count

A culture is diluted in tenfold steps: 1.0 cm^3 into 9.0 cm^3 of sterile water gives a 10^{-1} dilution, 1.0 cm^3 of that into another 9.0 cm^3 gives 10^{-2} , and so on. A known volume of each is spread on agar, and only plates with roughly 30–300 separate colonies are counted: fewer gives large random error, more and colonies merge.

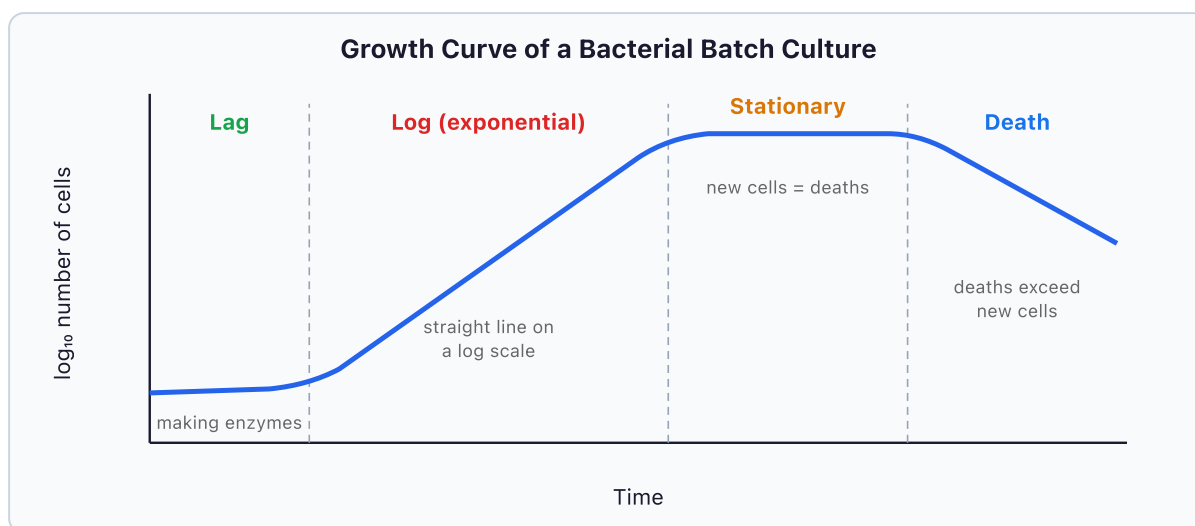
$$\text{Cells per cm}^3 \text{ of original culture} = \text{number of colonies} \times \text{dilution factor} \div \text{volume plated (cm}^3\text{)}$$

Worked example: A plate spread with 0.1 cm^3 of the 10^{-5} dilution grows 47 colonies. The dilution factor is 10^5 , so the original culture contained $47 \times 10^5 \div 0.1 = 4.7 \times 10^7$ living cells per cm^3 .



6.1 (v) The Bacterial Growth Curve

When a small number of bacteria are added to fresh broth in a closed flask, a plot of the logarithm of cell number against time shows four phases. A log scale is used because the numbers rise over several orders of magnitude; on a log scale, exponential growth appears as a straight line.



Phase	What is happening	Explanation
Lag phase	Little or no increase in cell number	Cells are adjusting to the new medium: taking up water, switching on genes and synthesising the enzymes needed to use the nutrients present. They are metabolically active and may grow in size, but divide little.
Log (exponential) phase	Cell number doubles at a constant rate	Nutrients are plentiful and waste is low, so cells divide at the fastest rate the conditions allow. The number of cells doubles every generation time.
Stationary phase	Population stays roughly constant	Nutrients are running low and toxic waste products (and in aerobes, a lack of oxygen) are building up. The rate of cell division equals the rate of cell death.
Death (decline) phase	Number of living cells falls	Nutrients are exhausted and waste has reached toxic levels, so the death rate is greater than the rate of division.

Exponential Growth and the Growth Rate Constant

Because each cell divides into two, a population starting with N_0 cells will contain $N_0 \times 2^n$ cells after n generations. For example, 500 cells that go through 6 generations become $500 \times 2^6 = 32,000$ cells. This is why a few cells can become millions within hours.

The **mean growth rate constant**, k , is the number of generations (doublings) per unit time during the log phase. Taking logarithms of $N_t = N_0 \times 2^{kt}$ gives:

$$k = (\log_{10} N_t - \log_{10} N_0) \div (\log_{10} 2 \times t) = (\log_{10} N_t - \log_{10} N_0) \div (0.301 \times t)$$

Here N_0 is the number of cells at the start of the time interval, N_t the number at the end, and t the length of the interval. The **mean generation (doubling) time** is $1 \div k$.

Worked example: During the log phase a culture rises from 5.0×10^2 to 3.2×10^4 cells cm^{-3} in 3 hours. $\log_{10}(3.2 \times 10^4) - \log_{10}(5.0 \times 10^2) = 4.505 - 2.699 = 1.806$. So $k = 1.806 \div (0.301 \times 3) = 2.0$ generations per hour, and the mean generation time is $1 \div 2.0 = 0.5$ hours, or 30 minutes.

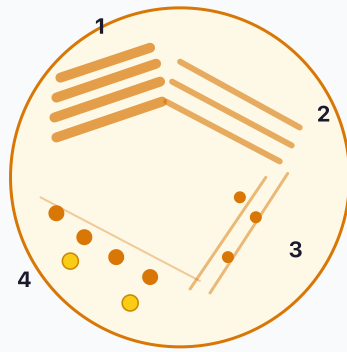
Exam Tip: Always use values from the straight (log) part of the curve when calculating k , and give the units (generations per hour, or h^{-1}). A common slip is to forget the 0.301 ($\log_{10} 2$): without it you have calculated the change in $\log_{10} N$ per hour, not the number of generations per hour. Another is reading the y-axis as cell number when it is actually $\log_{10}$ of cell number. If you are asked to *estimate* a population after a set time, check your answer is sensible by counting doublings.

Core Practical 12 (CP12): Measuring how fast bacteria grow in a broth culture, handling the organisms safely and ethically. Samples are taken at intervals and cells are measured by turbidity or dilution plating; $\log_{10}$ cell number is plotted against time. See the [core practicals page, CP12](#), for the full method, safety points and sources of error.

Isolating a Single Species: Streak Plating

To identify or study one species from a mixed sample, a **pure culture** is needed: a population grown from a single cell. Streak plating thins out cells across an agar plate. After each area is streaked, the loop is flamed and cooled and then dragged from the previous area into a fresh one, carrying over fewer cells each time, until cells in the last area are far enough apart to grow into separate colonies.

Streak Plate After Incubation



Area 1: dense, confluent growth

Loop flamed and cooled between areas

Each area dragged from the previous one, so fewer cells are carried over

Area 4: separate colonies; two colour types show two species in the mixture

Pick one colony to start a pure culture

Core Practical 13 (CP13): Separating single species out of a mixed bacterial culture by streak plating, again with safe and ethical handling. See the [core practicals page, CP13](#), for the method, safety points and sources of error.

Exam Tip: For "safe and ethical use of organisms" in CP12 and CP13, think about non-pathogenic strains, aseptic technique, incubating at 25 °C or below, not opening plates after incubation, and killing all cultures by autoclaving before disposal. The ethical side is that cultures must not be released into the environment and that living material should not be used wastefully.

6.2 Bacteria as Pathogens

6.2 (i) How Bacteria Cause Disease

Pathogen: A microorganism (or virus) that causes disease. A pathogenic bacterium must enter the host, avoid the host's defences long enough to multiply, and then cause harm, either by damaging tissues directly or by releasing toxins.

Bacteria enter by being inhaled or swallowed, or through cuts, bites and mucous membranes. They then cause disease by **invading and destroying host tissue** (multiplying in or between cells, sometimes secreting enzymes that digest tissue, with the host's inflammatory response adding to the damage) and by **producing toxins** that interfere with host cells, sometimes far from where the bacteria are growing.

6.2 (ii) Exotoxins, Endotoxins and Tissue Invasion

Feature	Exotoxins	Endotoxins
Chemical nature	Usually proteins	Lipopolysaccharides (LPS): lipid and polysaccharide molecules in the outer membrane of Gram-negative bacteria
How released	Secreted by living bacteria into their surroundings	Part of the cell wall; released mainly when bacterial cells die and break apart
Effect	Often very specific, acting on particular cells or membranes; many are highly potent	General effects, such as fever, inflammation and, in large amounts, a dangerous fall in blood pressure (shock)
Heat stability	Usually destroyed by heating (proteins denature)	Relatively heat-stable
Spec example	<i>Staphylococcus</i> spp.	<i>Salmonella</i> spp.

***Staphylococcus* (exotoxins).** *Staphylococcus aureus* is often carried harmlessly on the skin and in the nose, but causes disease when it gets into wounds, food or the blood. Different strains secrete different protein exotoxins: some damage cell surface membranes so that host cells burst; enterotoxins made by bacteria growing in food cause vomiting and diarrhoea within hours of eating; others cause toxic shock syndrome or make the outer skin layers peel away (scalded skin syndrome).

***Salmonella* (endotoxins).** *Salmonella* species are Gram-negative bacteria usually taken in with contaminated food, such as undercooked poultry or eggs. They invade the cells lining the intestine. The lipopolysaccharide in their outer membrane acts as an endotoxin: when bacteria die and release it, it triggers a strong inflammatory response in the gut wall. This produces diarrhoea, abdominal pain and fever. Food poisoning salmonellae usually stay in the gut, whereas the typhoid species, *Salmonella* Typhi, spreads through the body.

***Mycobacterium tuberculosis* (invasion of host tissue).** TB bacteria are inhaled in droplets from an infected person's cough. Alveolar macrophages engulf them, but their thick, waxy cell wall resists digestion, so they survive and multiply inside the macrophages. The immune system walls off infected areas in tubercles (granulomas), and the infection may stay dormant for years. If immunity is weakened, for example by HIV or malnutrition, the bacteria become active, destroying lung tissue and causing

cavities, a persistent cough that may bring up blood, weight loss and fever; they can spread in the blood to other organs.

Exam Tip: Do not write that endotoxins are "released inside the body" and exotoxins "outside": both act inside the host. The distinction is whether the toxin is *secreted* by living cells (exo-) or is part of the cell wall released when cells *die* (endo-). A useful consequence: antibiotics that kill Gram-negative bacteria can briefly make symptoms worse, because dying cells release more endotoxin.

6.3 Action of Antibiotics

Antibiotic: A chemical, originally produced by a microorganism (many are now modified or made synthetically), that kills bacteria or stops them growing, while doing little harm to the host's cells. Antibiotics do not act against viruses.

Antibiotics can be used against bacteria because they target structures or processes that differ between bacterial and human cells. Bacteria have peptidoglycan cell walls (human cells have none) and 70S ribosomes (human cytoplasm has 80S ribosomes). Antibiotics are grouped by their effect:

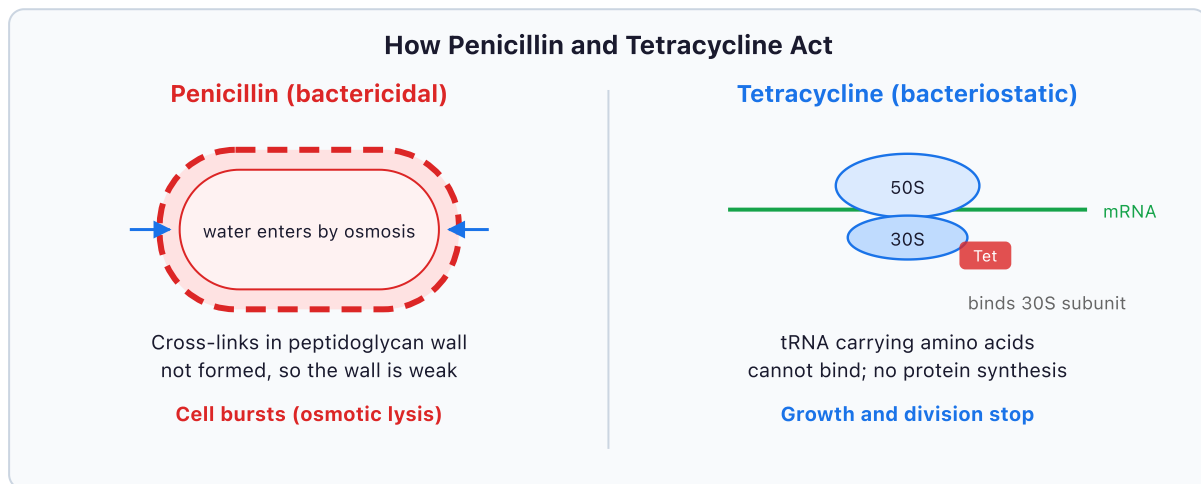
- **Bactericidal** antibiotics kill bacteria.
- **Bacteriostatic** antibiotics stop bacteria growing and dividing without killing them. The population stays in check while the immune system, especially phagocytes, destroys the bacteria. If the drug is stopped too early, or the patient's immune system is weak, the bacteria can start multiplying again.

Penicillin: Bactericidal

Penicillin inhibits the enzymes (transpeptidases, also called penicillin-binding proteins) that form the cross-links between peptidoglycan chains while a bacterium is building new cell wall. The bacterium keeps growing, but the new wall is weak. Water enters the cell by osmosis, because the cytoplasm has a lower water potential than the surroundings, and without a strong wall to resist the pressure the cell bursts (osmotic lysis). Penicillin therefore works best on bacteria that are actively growing and dividing, and is most effective against Gram-positive bacteria, whose thick peptidoglycan layer is exposed on the outside.

Tetracycline: Bacteriostatic

Tetracycline binds to the small (30S) subunit of the bacterial 70S ribosome and blocks the site where tRNA molecules carrying amino acids attach, so the polypeptide cannot be extended. Without protein synthesis the bacteria cannot make the enzymes and proteins needed to grow and divide, so the population stops increasing. It is broad-spectrum, and affects human cells much less because bacteria actively take up the drug and it binds 80S ribosomes less strongly.



Exam Tip: Be exact about penicillin. It does not "break down" or "dissolve" the cell wall; it stops cross-links forming in *new* wall, so the wall becomes weak as the cell grows, and water entering by osmosis bursts the cell. It also has little effect on bacteria that are not growing. For tetracycline, the key words are "binds to the 30S (small) ribosomal subunit" and "prevents tRNA binding", which "stops protein synthesis". If asked to *compare* the two, give a point about each drug in the same sentence.

6.4 Antibiotic Resistance

6.4 (i) Development and Spread of Resistance

Antibiotic resistance evolves by natural selection:

1. **Variation arises by mutation.** In a large population of bacteria, random mutations happen during DNA replication. By chance, a mutation may give a cell resistance, for example by producing an enzyme that breaks down the antibiotic (penicillinases, or β -lactamases, destroy penicillin), a pump

that removes the drug from the cell, or an altered target molecule that the drug can no longer bind to. The mutation is present before the antibiotic is used; the antibiotic does not cause it.

2. **The antibiotic is a selection pressure.** When the antibiotic is used, susceptible bacteria are killed or stop dividing. Resistant bacteria survive.
3. **Resistant bacteria reproduce.** With less competition, the survivors divide by binary fission and pass the resistance allele to their offspring (vertical transmission).
4. **The allele frequency increases.** Over many generations, which may be only 20–30 minutes each, the proportion of resistant bacteria in the population rises.

Resistance genes are often on **plasmids**, which can pass between bacteria, even of different species (horizontal gene transfer), mainly by **conjugation** through a connecting pilus. One plasmid may carry resistance to several antibiotics. Examples include **MRSA** (meticillin-resistant *Staphylococcus aureus*), *Clostridioides difficile* (which multiplies when broad-spectrum antibiotics kill a patient's normal gut bacteria) and multidrug-resistant *M. tuberculosis*.

Exam Tip: Never say bacteria "become resistant", "adapt to the antibiotic" or "build up immunity". Examiners penalise this. Say that a random mutation gave some bacteria resistance, the antibiotic killed the non-resistant ones, and the resistant ones survived, reproduced and passed on the allele, so the frequency of the allele increased. Bacteria do not have an immune system.

6.4 (ii) Controlling the Spread of Resistance, and Why It Is Difficult

Method	How it helps	Difficulties
Prescribe antibiotics only when needed, never for viral infections	Less use means less selection pressure	Patients may expect antibiotics; doctors may prescribe "just in case"; in some countries they can be bought without prescription
Identify the bacterium first, then use a narrow-spectrum antibiotic	Less selection pressure on other bacteria	Testing takes time, and a seriously ill patient may need treatment at once
Take the correct dose for the prescribed time	Reduces the chance that partly resistant bacteria	People stop when they feel better or keep leftover tablets

	survive	
Hospital infection control: hand hygiene, cleaning, screening and isolating patients	Stops resistant strains passing between patients	Many vulnerable patients close together; compliance is not always complete; isolation rooms are limited
Limit antibiotic use in farming	Routine use in animals selects for resistant bacteria that can reach humans	Farmers use them to prevent disease and boost growth; rules vary between countries
Develop new antibiotics	Bacteria have not yet met them	Slow and expensive, and new drugs would be held in reserve, so companies have little financial incentive

International travel, rapid bacterial reproduction and horizontal gene transfer make control harder still.

6.5 Other Pathogenic Agents

The specification requires three further pathogens: a fungus, a virus and a protoctist. For each you need how it is transmitted, how it gets into and infects the host (mode of infection) and the harm it causes. Detailed life cycles are not needed.

Stem Rust of Wheat (*Puccinia graminis*)

Transmission: reddish-brown spores are produced in huge numbers from pustules on stems and leaves and carried long distances by wind (also by rain splash and on clothing or machinery), so the disease spreads between fields, regions and countries. **Mode of infection:** on a wet plant surface (such as dew) a spore germinates; the germ tube grows to a stoma and a hypha enters through the pore, then grows between the cells and forms haustoria that penetrate cell walls and absorb sugars and other nutrients.

Pathogenic effect: sugars that would have gone to the grain feed the fungus instead; pustules rupture the epidermis, increasing water loss; less tissue photosynthesises; weakened stems collapse (lodging); grain is small and shrivelled, and severe outbreaks can destroy a crop. Resistant wheat varieties and fungicides are used for control, but new rust races can overcome resistance genes, as the Ug99 race, identified in Uganda at the end of the 1990s, did.

Influenza Virus

Influenza has an RNA genome in a protein coat within a lipid envelope carrying two glycoproteins, haemagglutinin (H) and neuraminidase (N). **Transmission:** droplets and aerosols from coughs and sneezes, and contaminated surfaces. **Mode of infection:** haemagglutinin binds to receptors on the epithelial cells lining the airways; the virus enters and uses the cell's machinery to make viral RNA and proteins; new virus particles bud from the cell (neuraminidase helps release them) and infect neighbouring cells.

Pathogenic effect: infected cells die; loss of ciliated epithelium stops mucus being cleared, so bacteria can cause secondary pneumonia, which can be fatal in the elderly and vulnerable; cytokines from the immune response cause fever and aches. Frequent changes in surface antigens (antigenic drift by mutation, and occasionally antigenic shift when strains swap genes) explain repeated infections, the need for a new vaccine each year and occasional pandemics.

The Malarial Parasite (*Plasmodium*)

Malaria is caused by several *Plasmodium* species (protoctists); *P. falciparum* is the most dangerous.

Transmission: by a vector, the female *Anopheles* mosquito, which injects parasites in its saliva when taking a blood meal and picks them up again from an infected person's blood; also via infected blood and from mother to fetus. **Mode of infection:** parasites travel to the liver and multiply inside liver cells, then invade red blood cells, multiply, burst them and infect more. Living inside cells partly hides them from the immune system.

Pathogenic effect: red cells bursting together release parasite products that cause repeated bouts of fever, shivering and sweating; red cell destruction causes anaemia; in *P. falciparum*, infected cells stick in small vessels and reduce blood flow to organs, causing cerebral malaria, kidney failure and death, especially in young children and pregnant women.

Exam Tip: The mosquito is the **vector**, not the pathogen; *Plasmodium* is the pathogen. Only female *Anopheles* mosquitoes transmit malaria (they take blood meals to produce eggs). When you are asked about transmission, the mode of infection and the pathogenic effect, answer all three parts separately for the named organism: examiners often mark each part on its own.

6.6 Problems of Controlling Endemic Diseases

Endemic disease: A disease that is always present in a particular population or region, at a fairly steady level. Malaria is endemic in much of sub-Saharan Africa and parts of Asia and South America. (An *epidemic* is a sudden rise in cases above the usual level; a *pandemic* is an epidemic spread across many countries or continents.)

Malaria can be attacked at several points: the mosquito vector, contact between mosquitoes and people, and the parasite inside people. Each approach has social, economic and ethical implications.

Control method	How it works	Social, economic and ethical implications
Insecticide-treated bed nets	Keep night-biting mosquitoes off sleepers and kill those that land	Cheap and effective, but must be distributed, replaced and used every night; mosquitoes can become resistant to the insecticide
Indoor spraying of insecticides (including DDT in some places)	Kills mosquitoes resting on walls after feeding	Needs repeated spraying and trained staff; resistance develops; DDT persists and accumulates in food chains, so its use is controversial
Removing breeding sites; larvicides; larva-eating fish	Reduces mosquito numbers at the larval stage	Breeding sites are numerous and hard to find; draining wetlands harms other wildlife; needs community effort
Antimalarial drugs	Kill parasites in the blood or liver	Cost to poor health systems and families; <i>Plasmodium</i> has evolved resistance to drugs such as chloroquine, so combinations (such as artemisinin-based combination therapies) are used; fake drugs encourage resistance
Vaccination	Stimulates immunity to the parasite	Hard to develop because the parasite changes its antigens and hides inside cells. The WHO recommended the RTS,S (2021) and R21 (2023) vaccines for children in endemic areas; they need several doses and give partial protection, raising questions of cost, delivery and fair allocation

Genetically modified or sterile mosquitoes	Released males reduce breeding or spread genes that reduce transmission	Released organisms cannot be recalled; possible effects on food webs; need for community consent and regulation
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Wider issues. Malaria is most common in low-income countries and keeps them poorer, because sick people cannot work or attend school. Governments may lack money for health workers and drugs and depend on uncertain international funding; conflict and poor infrastructure make rural programmes hard to run. Ethical questions include who should pay, how to share scarce drugs and vaccines fairly, whether environmentally harmful chemicals are justified to save lives, and how to run trials in poor communities with properly informed consent.

The Role of the Scientific Community in Validating Control Methods

Before a method is adopted widely, scientists test it in large, controlled, ideally randomised trials (for example comparing malaria cases in villages with and without bed nets); publish the results after **peer review** by independent experts; discuss them at conferences; repeat the studies elsewhere to check they can be **reproduced**, since mosquitoes behave differently in different regions; combine the evidence in systematic reviews used by bodies such as the World Health Organization; and keep monitoring for resistance and side effects after a method is introduced.

Success is possible where methods are combined and sustained: Sri Lanka, for example, went from a very high malaria burden to being certified malaria-free by the WHO in 2016, using a mix of vector control, case detection and treatment.

Exam Tip: "Discuss" or "evaluate" questions on malaria control need both sides and a conclusion. For each method give a benefit and a problem, and include at least one social, one economic and one ethical point. A strong answer also notes that no single method is enough, and that methods must be validated by peer-reviewed trials before they are relied on.

6.7 Response to Infection

6.7 (i) Macrophages, Neutrophils and Lymphocytes

White blood cells are central to the body's response. Two groups are distinguished: phagocytes (neutrophils and macrophages), which engulf pathogens and respond to any invader in the same way, and lymphocytes (B cells and T cells), which respond specifically to particular antigens.

Cell	Features	Mode of action
Neutrophil	The most common white blood cell; lobed nucleus; short-lived	Arrives first at a site of infection, attracted by chemicals released by damaged tissue and pathogens. Engulfs pathogens by phagocytosis and digests them. Dies after doing so; dead neutrophils, bacteria and tissue fluid form pus.
Macrophage	Large; develops from monocytes that leave the blood and settle in tissues; long-lived	Engulfs and digests pathogens and debris. Keeps fragments of the pathogen's antigens and displays them on its surface bound to MHC proteins, becoming an antigen-presenting cell that activates T helper cells. Also releases cytokines that attract other immune cells.
Lymphocytes (B and T cells)	Small cells with a large nucleus. Each lymphocyte has receptors of one specific shape on its surface, complementary to one antigen. B cells mature in the bone marrow; T cells mature in the thymus.	Only the lymphocytes with receptors complementary to the invading antigen are activated. B cells produce antibodies (humoral response). T cells act against infected body cells and coordinate the response (cell-mediated response).

Phagocytosis: the phagocyte moves towards the pathogen along a chemical gradient (chemotaxis) and binds to it, which is easier if the pathogen is coated with antibodies (opsonisation); it engulfs the pathogen in a vesicle (phagosome); lysosomes fuse with the phagosome and their hydrolytic enzymes digest the pathogen.

6.7 (ii) The Humoral Immune Response

The humoral response is the production of antibodies, which circulate in the blood plasma and tissue fluid (the "humours"). It acts mainly against pathogens and toxins that are outside cells.

1. **Antigen presentation.** A macrophage engulfs a pathogen and displays its antigens bound to MHC proteins on its surface. This antigen-presenting cell activates a T helper cell with a complementary receptor. (The specification says "antigen presenting T cells"; the key idea is that presented antigen activates a matching T helper cell.)
2. **T helper cells and cytokines.** The T helper cell divides by mitosis and releases **cytokines** (such as interleukins) that stimulate other immune cells.
3. **B cells and clonal selection.** Only a few of the millions of different B cells have surface antibody receptors complementary to this antigen. These bind the antigen and, when a matching T helper cell binds and releases cytokines, are activated. Choosing the B cell with the right receptor in this way is **clonal selection**.
4. **Clonal expansion.** The selected B cell divides repeatedly by mitosis into a clone of identical cells specific to the same antigen.
5. **Plasma cells.** Most clone cells become plasma cells, rich in rough endoplasmic reticulum and Golgi, which secrete large amounts of one specific antibody for a few days to weeks.
6. **Memory cells.** Some become long-lived B memory cells (see 6.7 iv).

Antibody: A globular glycoprotein (immunoglobulin) made of four polypeptide chains, two heavy and two light, held together by disulfide bridges in a Y shape. The tips of the two arms form two identical **antigen-binding sites** (variable regions) whose shape is complementary to one specific antigen. The rest of the molecule (constant region) is the same in all antibodies of a class and binds to receptors on phagocytes.

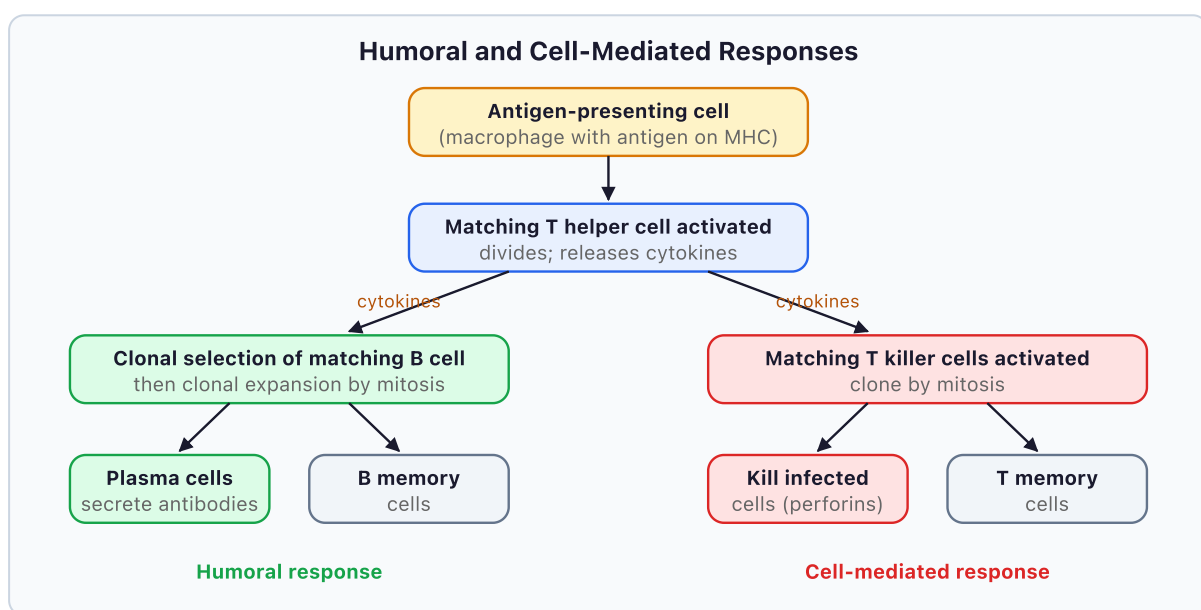
Antibodies neutralise toxins, block the sites pathogens use to attach to host cells, clump pathogens together (agglutination) because each has two binding sites, and act as markers (opsonins) for phagocytes.

6.7 (iii) The Cell-Mediated Immune Response

Pathogens such as viruses, and bacteria such as *M. tuberculosis*, spend much of their time inside body cells, where antibodies cannot reach them. The cell-mediated response destroys the infected cells themselves.

1. **Antigen-presenting cells.** Macrophages that have engulfed a pathogen present its antigens on their surface. Body cells infected with a virus also display fragments of viral protein on MHC proteins on their surface, marking them as infected.

2. **T helper cells and cytokines.** A T helper cell with a complementary receptor binds to the presented antigen, is activated and divides to form a clone. Its cytokines stimulate the matching T killer cells to divide, and also stimulate phagocytes.
3. **T killer cells.** T killer (cytotoxic) cells with receptors complementary to the antigen are activated and multiply by mitosis. They bind to infected body cells showing the same antigen and release proteins called **perforins**, which make pores in the target cell's membrane. Enzymes enter through the pores and trigger the cell to die (apoptosis), destroying the pathogens inside with it before they can multiply further.
4. Some activated T cells become T memory cells.



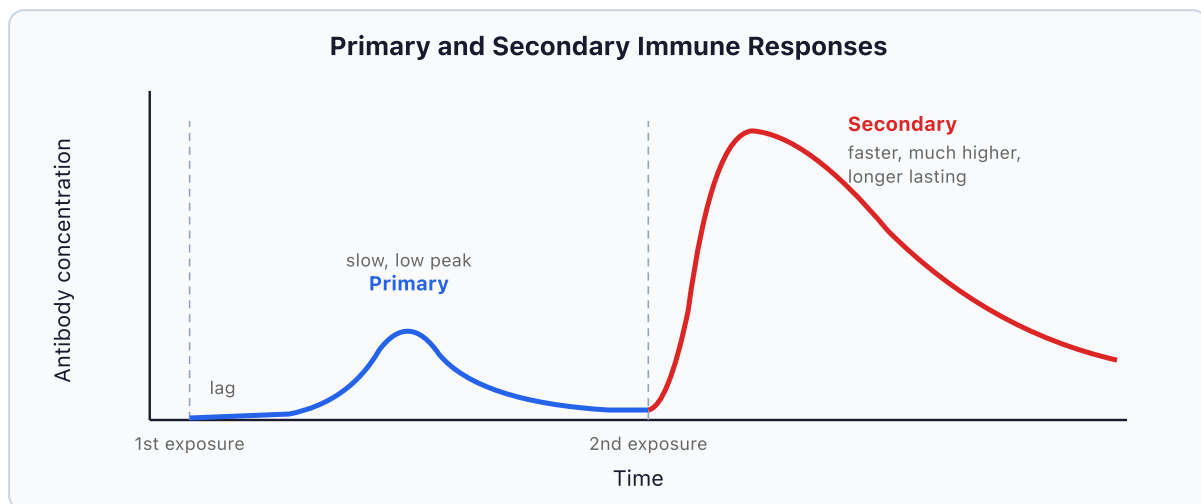
Exam Tip: Keep the cell types distinct. **Plasma cells**, not B cells in general and never T cells, secrete antibodies. **T helper cells** do not kill anything; they activate other cells by releasing cytokines. **T killer cells** kill infected *body cells*, not free pathogens. And antigens are on the pathogen, antibodies are made by the host: "antibodies on the surface of the bacterium" is a classic slip.

6.7 (iv) Memory Cells and the Secondary Immune Response

The first time the body meets an antigen, the **primary response** is slow: it takes days for the few matching lymphocytes to be selected, cloned and turned into plasma cells, so antibody levels rise slowly and stay fairly low. During this time the pathogen may multiply enough to cause illness. The

primary response leaves behind large numbers of long-lived **B memory cells** and **T memory cells** specific to that antigen.

If the same antigen enters the body again, the memory cells recognise it at once. B memory cells divide rapidly and differentiate into plasma cells; T memory cells rapidly produce T helper and T killer cells. This **secondary response** is faster, produces a much higher concentration of antibody and lasts longer, so the pathogen is usually destroyed before it causes symptoms. The person is said to be immune.



Exam Tip: To explain the shape of a secondary response graph, refer to memory cells: there are many more of them than the original matching lymphocytes, they are already specific to the antigen so clonal selection does not have to start again, and they divide and become plasma cells quickly. Avoid saying "the antibodies remember the pathogen": antibodies are broken down after a few weeks; it is the memory *cells* that persist.

6.7 (v) Natural and Artificial, Active and Passive Immunity

Active immunity is when a person's own immune system responds to an antigen, producing antibodies and memory cells; it takes time to develop but is long-lasting. **Passive immunity** is when a person is given ready-made antibodies produced by another individual; it is immediate but short-lived, because no memory cells are made and the antibodies are gradually broken down. Either can be **natural** (happening through normal life) or **artificial** (resulting from deliberate medical intervention).

	Natural	Artificial
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Active (own antibodies and memory cells; long-lasting)	Catching an infection and recovering from it	Vaccination with antigens
Passive (antibodies from elsewhere; immediate but temporary)	Antibodies crossing the placenta from mother to fetus, or passed to a baby in breast milk (especially colostrum)	Injection of antibodies, for example antitoxins after a snake bite or tetanus-prone wound, or antibodies given after exposure to rabies

6.7 (vi) Vaccination and Herd Immunity

Vaccination: Deliberately introducing antigens into the body, usually by injection or by mouth, to stimulate a primary immune response and the production of memory cells without causing the disease. A later infection with the same pathogen then meets a rapid secondary response.

Vaccines may contain killed or weakened (attenuated) pathogens, inactivated toxins, purified antigens, or genetic material that makes body cells produce the antigen. A later **booster** dose triggers a secondary response, increasing memory cells and antibody levels.

Herd immunity arises when so many people in a population are immune that a pathogen cannot spread easily: an infected person is unlikely to meet anyone who is susceptible, so chains of transmission die out. This protects people who cannot be vaccinated, such as newborn babies, people with weakened immune systems and those with allergies to vaccine components, as well as the small proportion in whom a vaccine does not work. The proportion of the population that needs to be immune depends on how infectious the disease is. For a disease in which each case would, in a fully susceptible population, infect R_0 others on average, the threshold is approximately:

$$\text{Proportion immune needed} \approx 1 - 1 / R_0$$

Measles is extremely infectious, with R_0 usually quoted as 12–18. That gives a threshold of about 92–94%, which is why coverage of around 95% is the target for measles vaccine. A disease with R_0 of 2 would need only about 50% immunity.

Vaccination works best where the pathogen's antigens are stable (unlike influenza), there is no animal reservoir, and the vaccine is cheap and easy to give so coverage can be high. Smallpox met all of these,

and was declared eradicated in 1980 after a worldwide vaccination campaign.

6.7 (vii) When Some People Choose Not to Vaccinate

If the proportion of people who are vaccinated falls below the herd immunity threshold, the unvaccinated can no longer rely on others to protect them. Susceptible people build up in the population, particularly children, and outbreaks can occur. The people most at risk include those too young to be vaccinated and those who cannot be vaccinated for medical reasons, who did not choose to be unprotected. Diseases that had become rare can return, with costs to health services for treatment and outbreak control.

For example, after a 1998 paper claimed a link between the MMR vaccine and autism, MMR uptake in England fell to roughly 80% in the early 2000s, well below the measles threshold, and measles cases rose in later years. Large studies found no link and the paper was retracted in 2010, showing the importance of replication and of communicating evidence clearly.

Ethical questions follow. Should vaccination be compulsory, or should individual choice be respected? Parents decide for children who cannot consent. Rare side effects have to be weighed against the much greater risks of the disease and the benefit to others through herd immunity.

Exam Tip: In a vaccination question, name the correct type of immunity: vaccination gives **artificial active** immunity. Injected antibodies give artificial passive immunity. Also be clear that vaccinated people are protected by their own memory cells, while herd immunity protects the unvaccinated by making it unlikely they will be exposed. Command words matter here: "explain" needs reasons (why coverage must exceed a threshold), while "discuss" needs points for and against (for example, compulsory vaccination).

Command Words

Command word	What it asks for in this topic
State / Give / Name	A short answer with no explanation, for example "Name the bacterium used as an example of an endotoxin producer" (<i>Salmonella</i>).

Describe	An account of what happens or what a graph shows, without reasons, such as the shape of a growth curve in each phase, with data.
Explain	Reasons linked to biology: why cell numbers level off in the stationary phase; why penicillin causes lysis.
Calculate	A numerical answer with working and units: a viable count, a growth rate constant, a herd immunity threshold.
Compare	Similarities and/or differences between two things, with a point about each: exotoxins and endotoxins; humoral and cell-mediated responses; primary and secondary responses.
Suggest	Apply your knowledge to an unfamiliar situation, such as why a new selective medium could isolate a particular pathogen.
Discuss / Evaluate	Advantages and disadvantages, or evidence for and against, leading to a judgement, as with malaria control methods or compulsory vaccination.

Key Terms Checklist

- ☐ Aseptic technique
- ☐ Autoclave
- ☐ Broth, agar and selective media
- ☐ Batch culture
- ☐ Viable count / total count
- ☐ Serial dilution
- ☐ Turbidity
- ☐ Lag, log, stationary and death phases
- ☐ Growth rate constant (k)
- ☐ Generation time
- ☐ Streak plating / pure culture
- ☐ Pathogen

- ☐ Exotoxin / endotoxin
- ☐ Bactericidal / bacteriostatic
- ☐ Antibiotic resistance
- ☐ Plasmid / conjugation
- ☐ MRSA
- ☐ Vector
- ☐ Haustorium
- ☐ Endemic
- ☐ Peer review
- ☐ Neutrophil / macrophage
- ☐ Antigen / antibody
- ☐ Antigen-presenting cell
- ☐ Cytokines
- ☐ Clonal selection / clonal expansion
- ☐ Plasma cell
- ☐ T helper cell / T killer cell
- ☐ Humoral / cell-mediated response
- ☐ Memory cells
- ☐ Active / passive, natural / artificial immunity
- ☐ Vaccination
- ☐ Herd immunity